

Saurabh Rathore

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— PROFILE

Physical oceanographer and climate scientist working on how the ocean stores and moves heat, drives sea-level rise and extremes, and shapes a changing climate. Completed a PhD at the Institute for Marine and Antarctic Studies (University of Tasmania), in collaboration with CSIRO and the Woods Hole Oceanographic Institution (WHOI), on global ocean heat-content change and the use of sea-surface salinity as a precursor for Australian rainfall prediction with machine learning. Held postdoctoral positions at LOCEAN (Sorbonne University, Paris) and the Climate Change Research Centre (UNSW, Sydney), with additional research experience at WHOI, IMAS, IITM Pune, and IIT Kharagpur. Research spans ocean warming, sea-level rise, marine heatwaves, ocean circulation, eddy dynamics, and ocean–climate modelling.

— RESEARCH INTERESTS

Physical oceanography & ocean circulation

Ocean heat uptake & sea-level rise

Marine heatwaves

Compound extreme events

Air–sea interactions & teleconnections

Global & regional hydrological cycle

Detection & attribution

Ocean eddy dynamics

Antarctic Sea Ice

Numerical ocean modelling

Data analysis & machine learning

— ACADEMIC APPOINTMENTS

Assistant Professor

Centre for Atmospheric Sciences, IIT Delhi

Aug 2023 – Present

Postdoctoral Fellow

ACEAS, Climate Change Research Centre (CCRC), UNSW, Sydney

Detection and attribution of sea-level rise.

Sep 2022 – Jul 2023

Postdoctoral Fellow

LOCEAN, Sorbonne University, Paris

Physical oceanography and ocean modelling.

Feb 2021 – Aug 2022

Casual Researcher

UNSW, Sydney — with Dr Andrea Taschetto

Atmospheric teleconnections to ENSO for Australia and South America.

Nov – Dec 2020

— EDUCATION

PhD, Physical Oceanography & Climate Science

Institute for Marine and Antarctic Studies (IMAS), University of Tasmania, Hobart

In collaboration with CSIRO (Australia) and WHOI (USA).

2021

Graduate Diploma of Marine & Antarctic Science

Institute for Marine and Antarctic Studies (IMAS), Hobart

2021

M.Tech, Earth System Science & Technology

Indian Institute of Technology Kharagpur (IIT KGP)

2015

B.E. Equivalent, Aerospace Engineering (AMAESEI)

The Aeronautical Society of India (AeSI), New Delhi

2013

--- **PUBLICATIONS**

1. **Rathore, S.**, Bindoff, N. L., Phillips, H. E., et al. (2020). Recent hemispheric asymmetry in global ocean warming induced by climate change and internal variability. *Nature Communications*, 11, 2008. doi:10.1038/s41467-020-15754-3
2. **Rathore, S.**, Goyal, R., Jangir, B., Ummenhofer, C. C., Feng, M., & Mishra, M. (2022). Interactions between a marine heatwave and Tropical Cyclone Amphan in the Bay of Bengal in 2020. *Frontiers in Climate*. doi:10.3389/fclim.2022.861477
3. **Rathore, S.**, Bindoff, N. L., Ummenhofer, C. C., Phillips, H. E., & Feng, M. (2020). Near-surface salinity reveals the oceanic sources of moisture for Australian precipitation through atmospheric moisture transport. *Journal of Climate*, 33(15). doi:10.1175/JCLI-D-19-0579.1
4. **Rathore, S.**, Bindoff, N. L., Ummenhofer, C. C., Phillips, H. E., Feng, M., & Mishra, M. (2021). Improving Australian rainfall prediction using sea surface salinity. *Journal of Climate*, 34(7). doi:10.1175/JCLI-D-20-0625.1
5. de Lavergne, C., **Rathore, S.**, Madec, G., Sallée, J.-B., Ethé, C., Nasser, A., et al. (2026). Effects of improved tidal mixing in the NEMO one-degree global ocean model. *Journal of Advances in Modeling Earth Systems*, 18, e2024MS004824. doi:10.1029/2024MS004824
6. Lyu, Y., Bindoff, N. L., Mohapatra, S., **Rathore, S.**, & Phillips, H. E. (2025). Global water cycle pattern amplification: contributing factors and mechanisms. *Journal of Geophysical Research: Oceans*, 130, e2024JC022278. doi:10.1029/2024JC022278
7. Sallée, J.-B., et al. (2023). Southern Ocean carbon and heat impact on climate. *Philosophical Transactions of the Royal Society A*, 381, 20220056. doi:10.1098/rsta.2022.0056
8. Bindoff, N. L., et al. (2019). Changing Ocean, Marine Ecosystems, and Dependent Communities. In *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC)*.
9. **Rathore, S.**, et al. (2021). Using sea surface salinity to foretell Australian rainfall (feature article). *Australian Marine Sciences Association (AMSA) Bulletin*.

--- **INVITED & VISITING TALKS**

- “What Causes the Rapid and Intense Warming of the Southern Ocean: The Fingerprint of Asymmetric Internal Variability” — IIT Delhi (Oct 2021); IIT Bhubaneswar & NISER (Nov 2021).
- “Recent hemispheric asymmetry in global ocean warming...” — IIT Kharagpur (Feb 2020).
- NESP Science Webinar — “Ocean Heat Uptake in a Changing Climate” (Feb 2018).

--- **SELECTED CONFERENCE & WORKSHOP PRESENTATIONS**

- **AMOS 2021** — “A New Mode of Asymmetric Internal Variability in the Ocean” (oral); “Improving Australian Rainfall Prediction Using Sea Surface Salinity” (oral).
- **AGU Ocean Sciences Meeting 2020**, San Diego — ocean warming asymmetry; precipitation prediction from sea surface salinity (posters).
- **EGU 2019**, Vienna — “Links between Sea Surface Salinity and Terrestrial Precipitation” (poster; selected among top 3–5% for public & media interest).
- **AGU Ocean Sciences Meeting 2018**, Portland — “Hemispheric Asymmetry in Upper Ocean Heat Content” (poster).
- **AMOS-ICSHMO 2018**, Sydney — “Indian Ocean Dominance in Upper Ocean Heat Uptake since 2005” (oral).
- **Workshops** — CLEX Annual Workshop 2018; NESP ESCC Symposium 2018; ARCCSS Annual Workshop 2017. Earlier: RAROM 2016 (IIT KGP); OSICON 2015, Goa.

--- **RESEARCH VOYAGES & FIELDWORK**

- **RV Investigator, Voyage V2018_V05** (Oct–Nov 2018) — 32-day oceanographic research voyage: “How does a standing meander southeast of Tasmania brake the Antarctic Circumpolar Current?” Student investigator; VMP deployment and CTD water sampling.

— AWARDS & SCHOLARSHIPS

- UTAS PhD Scholarship in Physical Oceanography & Climate Change (2016); fully funded by University of Tasmania & CSIRO.
- M.Tech scholarship, MHRD India — All India Rank 59, GATE 2012 (Aerospace Engineering) (2013).
- QMS Travel Scholarship (EGU 2019); ARCCSS Travel Scholarships (WHOI visit 2018; OSM 2018 & AMOS 2018).
- EGU 2019 poster selected among the top 3–5% of abstracts for public & media interest.
- Marine National Facility (MNF) award, Voyage V2018_V05 (2018); supervisory travel support for AMOS 2021 and AGU OSM 2020.

— PROFESSIONAL SERVICE

- Reviewer — **Geophysical Research Letters** (AGU) and **Journal of Geophysical Research: Oceans** (AGU).
- Expert Reviewer — **IPCC AR6**, Working Group I, Second Order Draft.

— RESEARCH EXPERIENCE & INTERSHIPS

- **Woods Hole Oceanographic Institution (WHOI), USA** (Feb–Jun 2019) — with Dr Caroline Ummenhofer: sea surface salinity as a precursor of Australian precipitation.
- **IIT Kharagpur** (Jun 2015–Oct 2016) — with Dr Mihir Dash: Indo-Pacific teleconnections related to Indian Ocean warming.
- **IITM Pune** (May 2014–Mar 2015) — with Dr Anant Parekh: mesoscale eddy dynamics in the Bay of Bengal and their atmospheric impact.

— MEDIA & PUBLIC OUTREACH

- **The Hindu** — “How marine heatwave fuelled super cyclone Amphan.”
- **Down To Earth** — “Warmer subsurface waters in Bay of Bengal likely fuelled Amphan”; “Oceans' Great Dying 2.0: Earth's climate moderator is warming faster.”
- **Mongabay** — “Understanding cyclone behaviour in a changing climate.”
- Ocean-current simulation demonstrations for school students & the public — Taroona High School, Hobart (2019); Festival of Bright Ideas, Hobart (2017).

— TECHNICAL SKILLS

Programming & analysis: Python, MATLAB, Py-Ferret (NOAA PMEL), NCL (NCAR Command Language), CDO (Climate Data Operator), NCO (NetCDF Operator), NetCDF, machine learning. **Computing:** High-Performance Computing, Linux, scientific writing. **Field:** VMP (Velocity Microstructure Profiler) deployment, CTD water sampling.

— TRAINING & COURSES

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| ● Modelling the Climate System — ARC CoE for Climate Extremes (2019). | ● Emerging Trends in Ocean Observations & Data Analysis — INCOIS (2016). |
| ● The Science of Climate Change — ARC CoE for Climate System Science (2017). | ● Global Spectral Modelling (ISWT) — IIT Kharagpur (2014). |